

# C++ Assignment: Polymorphic Shape Calculator

## OBJECT-ORIENTED PROGRAMMING & POLYMORPHISM

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### Objective

Implement a geometric shape tracking system using C++ inheritance and polymorphism. You will create an abstract base class and extend it to handle regular polygons with 3, 4, 5, and 6 sides.

### 1. Architectural Concept: The Base Class (Shape)

In C++, an Interface is simulated using an abstract class containing only pure virtual functions (`= 0`) and a virtual destructor.

The purpose of the `Shape` base class is to define a strict contract. Any specific shape added to the system must know how to calculate its own area and perimeter. By interacting only with the `Shape` interface, the main application can calculate metrics for a list of completely different shapes without knowing their specific types at compile time.

#### Core Interface Contract:

```
virtual double getArea() const = 0;
virtual double getPerimeter() const = 0;
virtual ~Shape(); // Crucial to ensure derived destructors clear memory properly
```

### 2. Geometric Requirements & Formulas

For this assignment, we assume all shapes are regular polygons (all sides are of equal length  $s$ ).

### 3 Sides: Equilateral Triangle

Perimeter:

$$P = 3s$$

Area (Derived from the standard triangle formula using the Pythagorean theorem for height):

$$A = (\sqrt{3} / 4) \times s^2$$

### 4 Sides: Square

Perimeter:

$$P = 4s$$

Area:

$$A = s^2$$

### 5 Sides: Regular Pentagon

Perimeter:

$$P = 5s$$

Area (Uses the perimeter and the apothem):

$$A = \frac{1}{4} \times \sqrt{5(5 + 2\sqrt{5})} \times s^2$$

*Tip: For code efficiency, compute the constant multiplier  $\frac{1}{4} \times \sqrt{5(5 + 2\sqrt{5})} \approx 1.720477$  once.*

### 6 Sides: Regular Hexagon

Perimeter:

$$P = 6s$$

Area (Composed of 6 equilateral triangles):

$$A = (3\sqrt{3} / 2) \times s^2$$

### 3. Student Deliverables

- **Define the Interface:** Implement the `Shape` abstract base class with the required pure virtual methods.
- **Implement the Derived Classes:** Create `Triangle`, `Square`, `Pentagon`, and `Hexagon`. Each class must accept the side length `s` in its constructor.
- **Enforce Safety:** Use the `override` keyword on all derived methods and ensure `const` correctness.
- **Polymorphic Test Driver:** In `main()`, instantiate a `std::vector<Shape*>` containing one instance of each shape. Loop through the vector to print out the Area and Perimeter of each entry dynamically, then properly clean up the heap allocations.